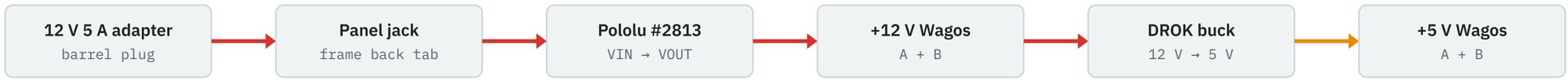
# Ball balancer wiring

Every connection, generated from the same list as the tables below (`docs/wiring.py`). Power and ground run through eight WAGO 221-415 lever connectors: two for +12 V, two for +5 V, four for ground, each pair or group linked by a short jumper. Signals are Dupont leads. Hover a wire to see its two ends.

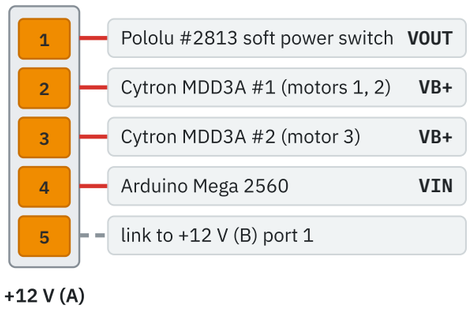
+12 V · +5 V · GND · Wago link · motor + · motor – · encoder A · encoder B · PWM · touch panel · buttons · power-off control

## Power: what plugs into every Wago port

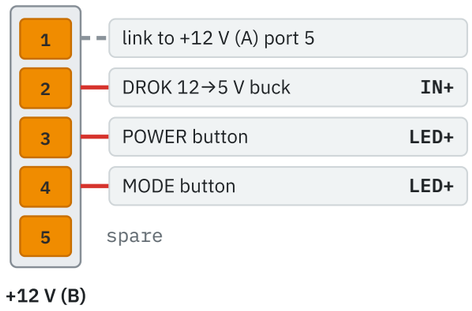



Every ground goes to the GND Wagos below. Linked Wagos act as one bus: the dashed port joins them.

### +12 V bus

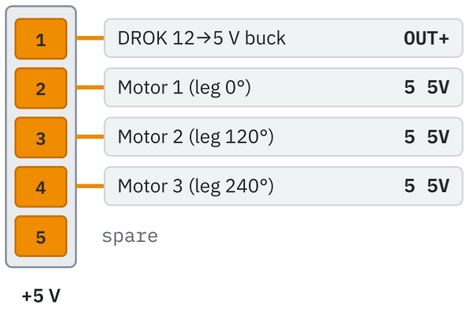



+12 V (A)



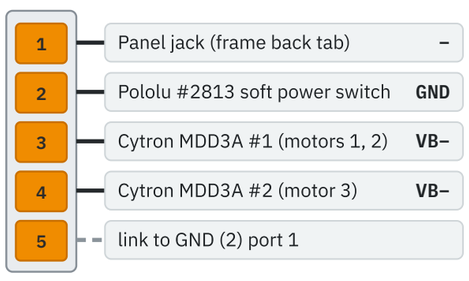

+12 V (B)

### +5 V bus

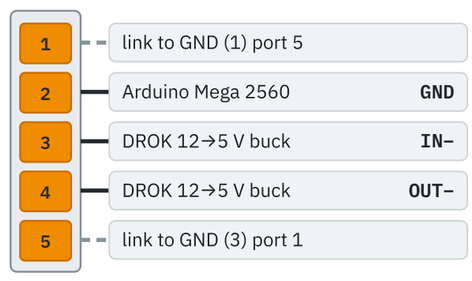



+5 V

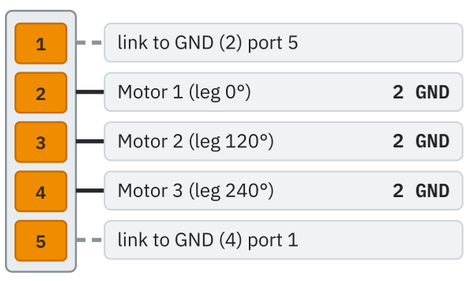
### Ground bus



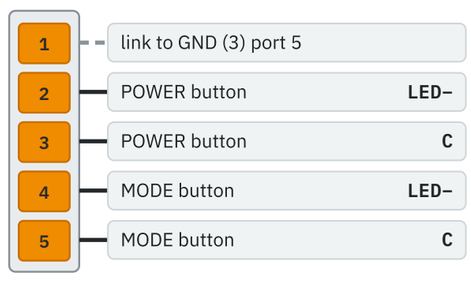

GND (1)




GND (2)

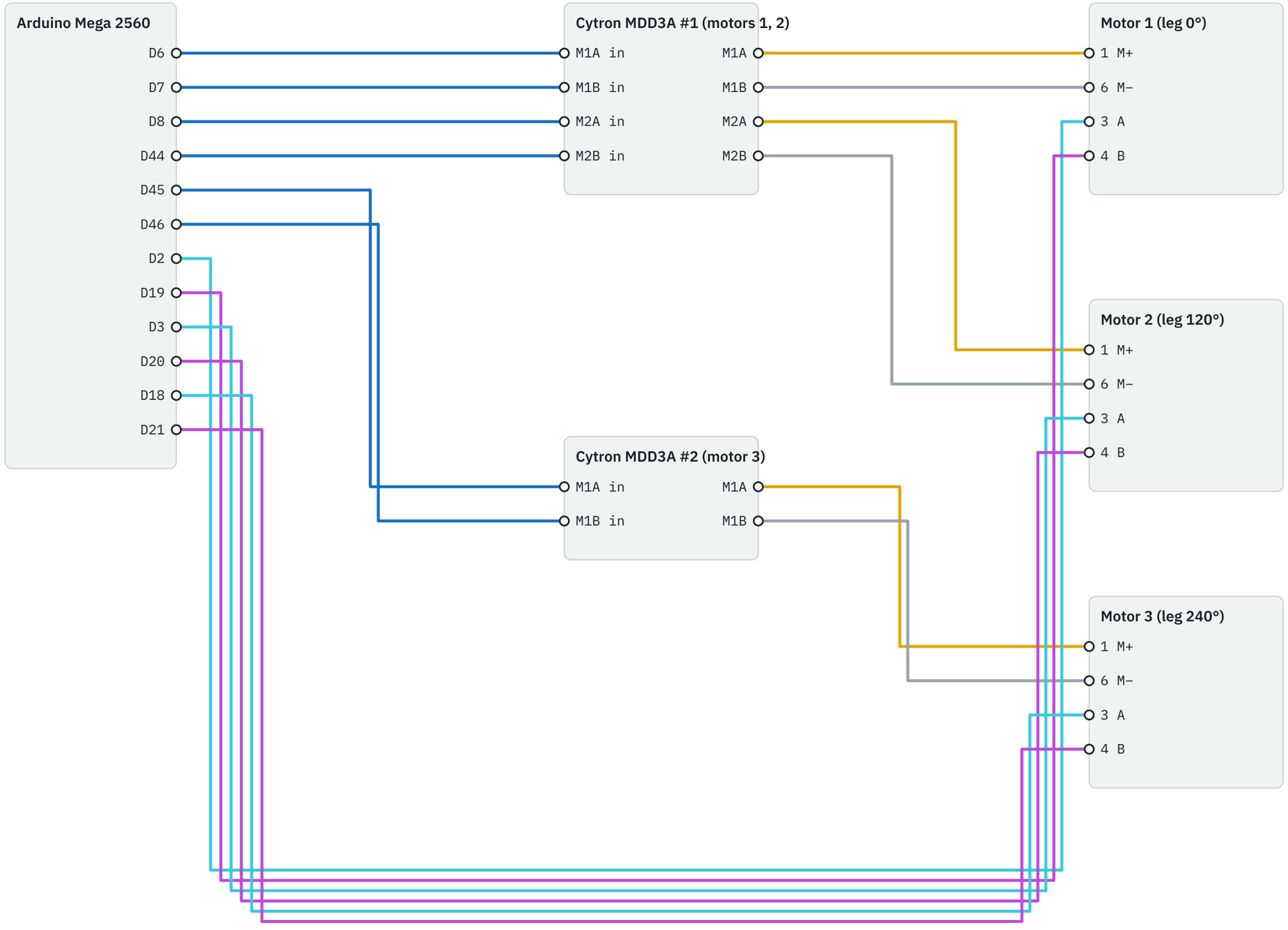



GND (3)

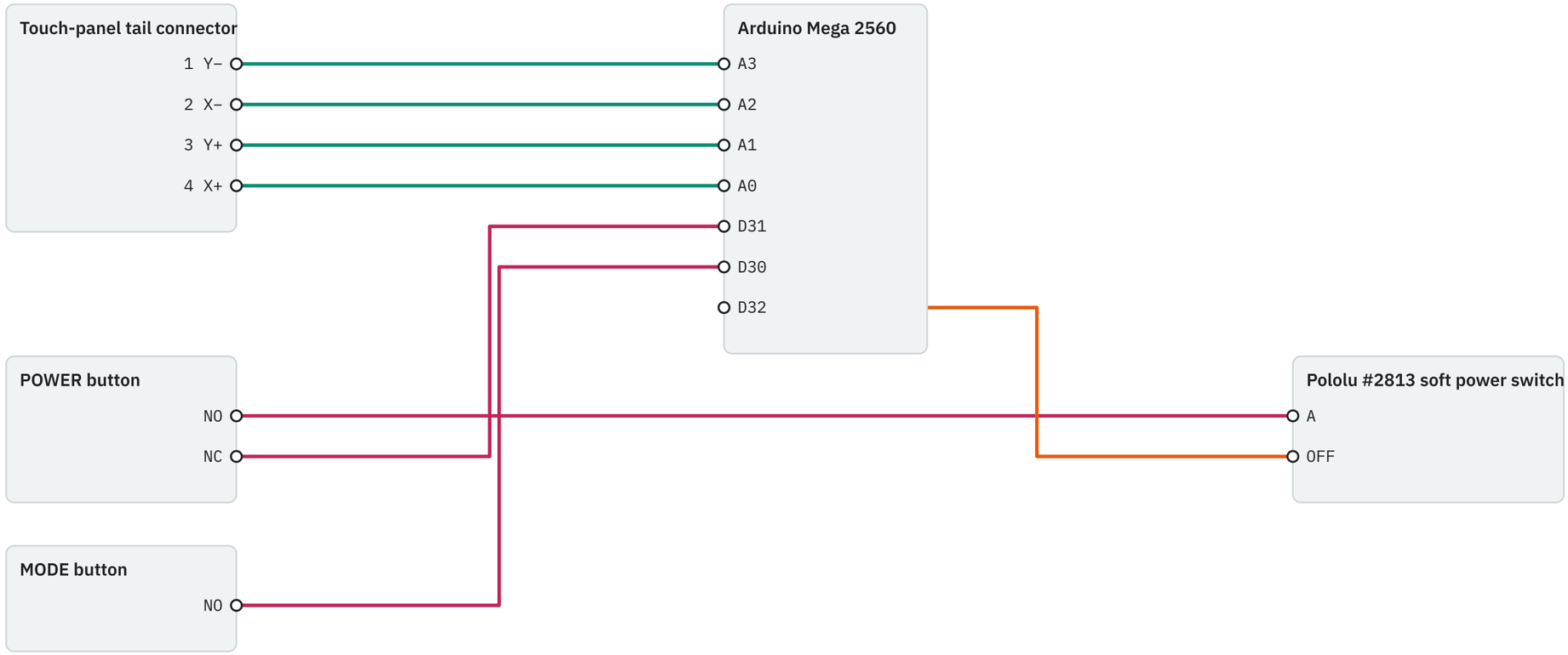



GND (4)

## Motors: control signals, motor power and encoders



## Touch panel, buttons and soft power




### Before you power it on

- Fit the motors' leads last. First power-up: no motors connected, check 12 V on the +12 V Wagos and 5.0 V on the +5 V Wagos with a meter.
- Never connect the buck's 5 V output and the Mega's USB 5 V together; the Mega takes 12 V on VIN only and makes its own 5 V.
- Touch-panel tail order (Y−, X−, Y+, X+) is from a similar panel's datasheet: confirm with a meter (about 200–900 Ω across each X pair and each Y pair).
- Motor connector pins, left to right: 1 M+, 2 GND, 3 A, 4 B, 5 5V, 6 M−. Port labels give the pin number.
- Our motor cable colours: M+ yellow, GND black, A blue, B purple, 5V red, M− white.
- Our button harness colours: NO blue, C green, NC yellow, LED+ red, LED− black.
- Button harness colours vary by batch: identify LED+, LED−, C, NO and NC with a meter before connecting.
- Program the Mega over USB with the shell lifted; the soft switch can stay off while USB powers the Mega's logic.
- Second MDD3A's channel 2 is unused. Leave both MDD3As' 5V pins unconnected: they are outputs from the board's own regulator.

## Wire list by part

### Pololu #2813 soft power switch

| PIN | GOES TO | WIRE |
|---|---|---|
| A | POWER button · NO | button harness to Dupont |
| GND | Wago GND (1) · 2 | black, 20 AWG |
| OFF | Arduino Mega 2560 · D32 | Dupont |
| VIN | Panel jack (frame back tab) · + | red, 20 AWG |
| VOUT | Wago +12 V (A) · 1 | red, 20 AWG |

### Panel jack (frame back tab)

| PIN | GOES TO | WIRE |
|---|---|---|
| + | Pololu #2813 soft power switch · VIN | red, 20 AWG |
| – | Wago GND (1) · 1 | black, 20 AWG |
| socket | 12 V 5 A wall adapter · plug | barrel plug |

### DROK 12→5 V buck

| PIN | GOES TO | WIRE |
|---|---|---|
| IN+ | Wago +12 V (B) · 2 | red, 20 AWG |
| IN– | Wago GND (2) · 3 | black, 20 AWG |
| OUT+ | Wago +5 V · 1 | red/orange, 20 AWG |
| OUT– | Wago GND (2) · 4 | black, 20 AWG |

### Arduino Mega 2560

| PIN | GOES TO | WIRE |
|---|---|---|
| A0 | Touch-panel tail connector · 4 X+ | Dupont |
| A1 | Touch-panel tail connector · 3 Y+ | Dupont |
| A2 | Touch-panel tail connector · 2 X– | Dupont |
| A3 | Touch-panel tail connector · 1 Y– | Dupont |
| D18 | Motor 3 (leg 240°) · 3 A | motor pin 3, A (blue) to Dupont |
| D19 | Motor 1 (leg 0°) · 4 B | motor pin 4, B (purple) to Dupont |
| D2 | Motor 1 (leg 0°) · 3 A | motor pin 3, A (blue) to Dupont |
| D20 | Motor 2 (leg 120°) · 4 B | motor pin 4, B (purple) to Dupont |
| D21 | Motor 3 (leg 240°) · 4 B | motor pin 4, B (purple) to Dupont |
| D3 | Motor 2 (leg 120°) · 3 A | motor pin 3, A (blue) to Dupont |
| D30 | MODE button · NO | button harness to Dupont |
| D31 | POWER button · NC | button harness to Dupont |
| D32 | Pololu #2813 soft power switch · OFF | Dupont |
| D44 | Cytron MDD3A #1 (motors 1, 2) · M2B in | Dupont |
| D45 | Cytron MDD3A #2 (motor 3) · M1A in | Dupont |
| D46 | Cytron MDD3A #2 (motor 3) · M1B in | Dupont |
| D6 | Cytron MDD3A #1 (motors 1, 2) · M1A in | Dupont |
| D7 | Cytron MDD3A #1 (motors 1, 2) · M1B in | Dupont |
| D8 | Cytron MDD3A #1 (motors 1, 2) · M2A in | Dupont |

### Cytron MDD3A #1 (motors 1, 2)

| PIN | GOES TO | WIRE |
|---|---|---|
| M1A | Motor 1 (leg 0°) · 1 M+ | motor pin 1, M+ (yellow) |
| M1B | Motor 1 (leg 0°) · 6 M– | motor pin 6, M– (white) |
| M2A | Motor 2 (leg 120°) · 1 M+ | motor pin 1, M+ (yellow) |
| M2B | Motor 2 (leg 120°) · 6 M– | motor pin 6, M– (white) |
| VB+ | Wago +12 V (A) · 2 | red, 20 AWG |
| VB– | Wago GND (1) · 3 | black, 20 AWG |
| M1A in | Arduino Mega 2560 · D6 | Dupont |
| M1B in | Arduino Mega 2560 · D7 | Dupont |
| M2A in | Arduino Mega 2560 · D8 | Dupont |
| M2B in | Arduino Mega 2560 · D44 | Dupont |

### Cytron MDD3A #2 (motor 3)

| PIN | GOES TO | WIRE |
|---|---|---|
| M1A | Motor 3 (leg 240°) · 1 M+ | motor pin 1, M+ (yellow) |
| M1B | Motor 3 (leg 240°) · 6 M– | motor pin 6, M– (white) |
| VB+ | Wago +12 V (A) · 3 | red, 20 AWG |
| VB– | Wago GND (1) · 4 | black, 20 AWG |
| M1A in | Arduino Mega 2560 · D45 | Dupont |
| M1B in | Arduino Mega 2560 · D46 | Dupont |

| PIN | GOES TO | WIRE |
|---|---|---|
| GND | Wago GND (2) · 2 | black, 20 AWG |
| VIN | Wago +12 V (A) · 4 | red, 20 AWG |

### Motor 1 (leg 0°)

| PIN | GOES TO | WIRE |
|---|---|---|
| 3 A | Arduino Mega 2560 · D2 | motor pin 3, A (blue) to Dupont |
| 4 B | Arduino Mega 2560 · D19 | motor pin 4, B (purple) to Dupont |
| 1 M+ | Cytron MDD3A #1 (motors 1, 2) · M1A | motor pin 1, M+ (yellow) |
| 2 GND | Wago GND (3) · 2 | black, 20 AWG |
| 5 5V | Wago +5 V · 2 | red/orange, 20 AWG |
| 6 M– | Cytron MDD3A #1 (motors 1, 2) · M1B | motor pin 6, M– (white) |

### Motor 2 (leg 120°)

| PIN | GOES TO | WIRE |
|---|---|---|
| 3 A | Arduino Mega 2560 · D3 | motor pin 3, A (blue) to Dupont |
| 4 B | Arduino Mega 2560 · D20 | motor pin 4, B (purple) to Dupont |
| 1 M+ | Cytron MDD3A #1 (motors 1, 2) · M2A | motor pin 1, M+ (yellow) |
| 2 GND | Wago GND (3) · 3 | black, 20 AWG |
| 5 5V | Wago +5 V · 3 | red/orange, 20 AWG |
| 6 M– | Cytron MDD3A #1 (motors 1, 2) · M2B | motor pin 6, M– (white) |

### Motor 3 (leg 240°)

| PIN | GOES TO | WIRE |
|---|---|---|
| 3 A | Arduino Mega 2560 · D18 | motor pin 3, A (blue) to Dupont |
| 4 B | Arduino Mega 2560 · D21 | motor pin 4, B (purple) to Dupont |
| 1 M+ | Cytron MDD3A #2 (motor 3) · M1A | motor pin 1, M+ (yellow) |
| 2 GND | Wago GND (3) · 4 | black, 20 AWG |
| 5 5V | Wago +5 V · 4 | red/orange, 20 AWG |
| 6 M– | Cytron MDD3A #2 (motor 3) · M1B | motor pin 6, M– (white) |

### Touch-panel tail connector

| PIN | GOES TO | WIRE |
|---|---|---|
| 1 Y– | Arduino Mega 2560 · A3 | Dupont |
| 2 X– | Arduino Mega 2560 · A2 | Dupont |
| 3 Y+ | Arduino Mega 2560 · A1 | Dupont |
| 4 X+ | Arduino Mega 2560 · A0 | Dupont |

### POWER button

| PIN | GOES TO | WIRE |
|---|---|---|
| C | Wago GND (4) · 3 | black, 20 AWG |
| NC | Arduino Mega 2560 · D31 | button harness to Dupont |
| NO | Pololu #2813 soft power switch · A | button harness to Dupont |
| LED+ | Wago +12 V (B) · 3 | red, 20 AWG |
| LED– | Wago GND (4) · 2 | black, 20 AWG |

### MODE button

| PIN | GOES TO | WIRE |
|---|---|---|
| C | Wago GND (4) · 5 | black, 20 AWG |
| NO | Arduino Mega 2560 · D30 | button harness to Dupont |
| LED+ | Wago +12 V (B) · 4 | red, 20 AWG |
| LED– | Wago GND (4) · 4 | black, 20 AWG |

### Wago +12 V (A)

| PIN | GOES TO | WIRE |
|---|---|---|
| 1 | Pololu #2813 soft power switch · VOUT | red, 20 AWG |
| 2 | Cytron MDD3A #1 (motors 1, 2) · VB+ | red, 20 AWG |
| 3 | Cytron MDD3A #2 (motor 3) · VB+ | red, 20 AWG |
| 4 | Arduino Mega 2560 · VIN | red, 20 AWG |
| 5 | Wago +12 V (B) · 1 | short jumper, same colour as its bus |

### Wago +12 V (B)

| PIN | GOES TO | WIRE |
|---|---|---|
| 1 | Wago +12 V (A) · 5 | short jumper, same colour as its bus |
| 2 | DROK 12→5 V buck · IN+ | red, 20 AWG |
| 3 | POWER button · LED+ | red, 20 AWG |
| 4 | MODE button · LED+ | red, 20 AWG |

### Wago +5 V

| PIN | GOES TO | WIRE |
|---|---|---|
| 1 | DROK 12→5 V buck · OUT+ | red/orange, 20 AWG |
| 2 | Motor 1 (leg 0°) · 5 5V | red/orange, 20 AWG |
| 3 | Motor 2 (leg 120°) · 5 5V | red/orange, 20 AWG |
| 4 | Motor 3 (leg 240°) · 5 5V | red/orange, 20 AWG |

### Wago GND (1)

| PIN | GOES TO | WIRE |
|---|---|---|
| 1 | Panel jack (frame back tab) · – | black, 20 AWG |
| 2 | Pololu #2813 soft power switch · GND | black, 20 AWG |
| 3 | Cytron MDD3A #1 (motors 1, 2) · VB– | black, 20 AWG |
| 4 | Cytron MDD3A #2 (motor 3) · VB– | black, 20 AWG |
| 5 | Wago GND (2) · 1 | short jumper, same colour as its bus |

### Wago GND (2)

| PIN | GOES TO | WIRE |
|---|---|---|
| 1 | Wago GND (1) · 5 | short jumper, same colour as its bus |
| 2 | Arduino Mega 2560 · GND | black, 20 AWG |
| 3 | DROK 12→5 V buck · IN– | black, 20 AWG |
| 4 | DROK 12→5 V buck · OUT– | black, 20 AWG |
| 5 | Wago GND (3) · 1 | short jumper, same colour as its bus |

### Wago GND (3)

| PIN | GOES TO | WIRE |
|---|---|---|
| 1 | Wago GND (2) · 5 | short jumper, same colour as its bus |
| 2 | Motor 1 (leg 0°) · 2 GND | black, 20 AWG |
| 3 | Motor 2 (leg 120°) · 2 GND | black, 20 AWG |
| 4 | Motor 3 (leg 240°) · 2 GND | black, 20 AWG |
| 5 | Wago GND (4) · 1 | short jumper, same colour as its bus |

### Wago GND (4)

| PIN | GOES TO | WIRE |
|---|---|---|
| 1 | Wago GND (3) · 5 | short jumper, same colour as its bus |
| 2 | POWER button · LED– | black, 20 AWG |
| 3 | POWER button · C | black, 20 AWG |
| 4 | MODE button · LED– | black, 20 AWG |
| 5 | MODE button · C | black, 20 AWG |